

FMN050 Numerical Analysis for E3

Solutions for Homework assignment 2

1. Integration of tabulated data. If we implement both the trapezoidal rule and Simpson's rule, we get a kind of error estimate of the approximation to the average temperature T . According to the trapezoidal rule and Simpson's rule $T = 568.4986$ respectively $T = 568.0919$. We conclude that an approximation is $T = 568$.

The MATLAB algorithm

```
function [Itrap,Isim]=s42
re=9.38;
r0=14.58;
N=11;
h=(r0-re)/(N-1);
r=linspace(re,r0,N)';
T=[338 423 474 506 557 573 601 622 651 661 671]';
F=f(r,T);
G=g(r);
Itrap=trapezoid(h,F)./trapezoid(h,G);
Isim=simpson(2*h,F)./simpson(2*h,G);
```

The functions needed by this algorithm:

```
function f=f(r,T)
f=T.*r;

function g=g(r)
g=r;

function I=trapezoid(h,f)
I=sum(h/2*(f(1:end-1)+f(2:end))));

function I=simpson(h,f)
I=sum(h/6*(f(1:2:end-2)+4*f(2:2:end-1)+f(3:2:end))));
```

2. Simpson quadrature.

Task 2a):

Simpson's rule can be derived by approximating the function $f(x)$ by a second order polynomial from the three points $(0, f(0))$, $(\frac{1}{2}, f(\frac{1}{2}))$ and $(1, f(1))$. The Lagrange polynomial will give simple calculations, but any second order polynomial interpolating these points will do (the resulting polynomial

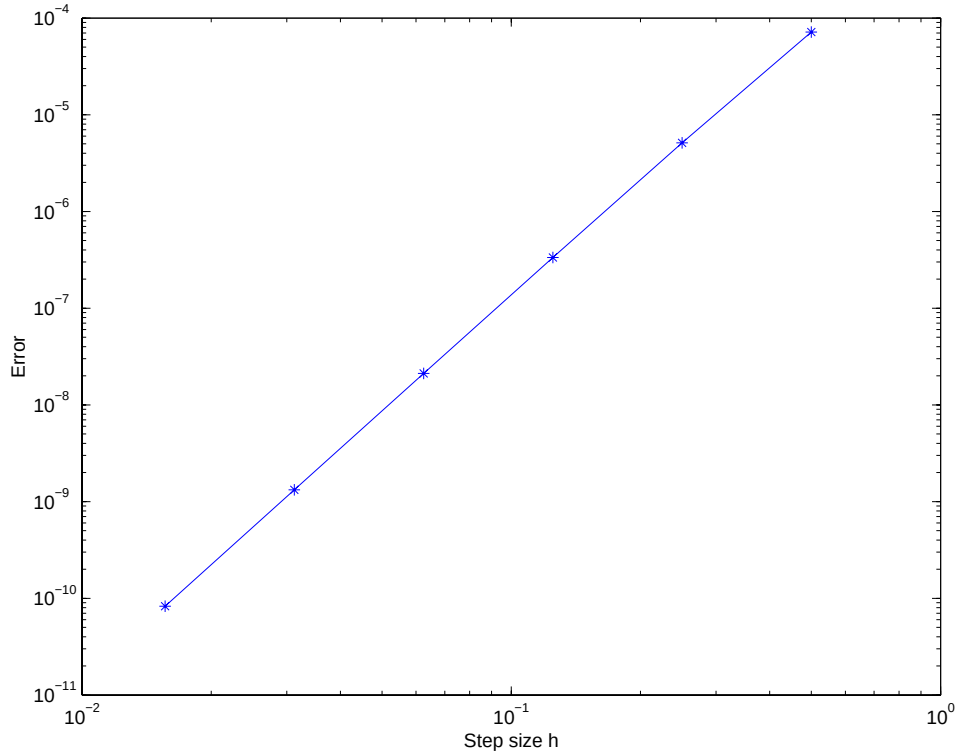


Figure 1: Error Vs. Step Size

is unique).

$$p_2(x) = f(0) \frac{(x - \frac{1}{2})(x - 1)}{(-\frac{1}{2})(-1)} + f\left(\frac{1}{2}\right) \frac{(x - 0)(x - 1)}{\frac{1}{2}(-\frac{1}{2})} + f(1) \frac{(x - 0)(x - \frac{1}{2})}{\frac{1}{2}}$$

$$p_2(x) = f(0) \cdot 2 \left(x - \frac{1}{2}\right) (x - 1) + f\left(\frac{1}{2}\right) \cdot (-4x(x - 1)) + f(1) \cdot 2x \left(x - \frac{1}{2}\right).$$

Integration of the expression above yields

$$\begin{aligned} f(0) \int_0^1 2(x - 1/2)(x - 1) dx + f(1/2) \int_0^1 -4x(x - 1) dx + f(1) \int_0^1 2x(x - 1) dx = \\ = \frac{1}{6} f(0) + \frac{2}{3} f(1/2) + \frac{1}{6} f(1) = \frac{1}{6} [f(0) + 4f(1/2) + f(1)]. \end{aligned}$$

This is exactly Simpson's rule.

Task 2b):

The M-file for this exercise is:

```

% generate w which is a 3-vector consisting of the weights for Simpson's rule.
w = [1 4 1]/6; I = 0.04934910257597;

% generate a vector for error and stepsize
err = zeros(1, 6); step = zeros(1, 6);

for k = 1:6
    h = 1/2^k;
    t = 0:h:0.5;
    n = length(t);
    Inum = 0;
    for i = 1:n-1
        x = linspace(t(i), t(i+1), 3);
        f = x.^2.*acos(x);
        Inum = Inum + h * w*f';
    end
    err(k) = abs(Inum - I);
    step(k) = h;
end plot(log10(step), log10(err), '*-')

```

Since we used a loglog scaled in Fig. 1, you can easily check the order of the method for this problem.

3. Gauss quadrature.

Task 3a):

The second order Legendre polynomial is $P_2(t) = \frac{1}{2}(3t^2 - 1)$. The knots c_j are given as the roots of

$$P_2(2c - 1) = \frac{1}{2} \left(3(2c - 1)^2 - 1 \right) = 0 \iff c_{\pm} = \frac{1}{2} \left(1 \pm \frac{1}{\sqrt{3}} \right). \quad (1)$$

In increasing order the knots are

$$c_1 = \frac{1}{2} - \frac{\sqrt{3}}{6}, \quad c_2 = \frac{1}{2} + \frac{\sqrt{3}}{6}. \quad (2)$$

The weights of a Gauss formula are uniquely determined by

$$b_j = \int_0^1 \mathcal{L}_j(\tau) d\tau, \quad (3)$$

where $\mathcal{L}_j$ is the Lagrange polynomial defined by $\mathcal{L}_j(c_i) = \delta_{ij}$ i.e.

$$\mathcal{L}_1(t) = \frac{t - c_2}{c_1 - c_2}. \quad (4)$$

Hence,

$$b_1 = \int_0^1 \mathcal{L}_1(t) dt = \frac{1}{c_1 - c_2} \left(\int_0^1 t dt - c_2 \right), \quad (5)$$

and

$$b_1 = \frac{1}{c_1 - c_2} \left(\frac{1}{2} - c_2 \right) = + \frac{3 \cdot \sqrt{3}}{\sqrt{3} \cdot 6} = \frac{1}{2}. \quad (6)$$

By consistency, $b_1 + b_2 = 1$, and the second weight is $b_2 = b_1 = 1/2$.

Remark: The weights b_j can also be determined by the order conditions (Theorem 32 in the compendium):

$$\sum_{j=1}^s b_j = 1, \quad \sum_{j=1}^s b_j c_j^{q-1} = \frac{1}{q}, \quad q = 1, 2, \dots, k, \quad (7)$$

where $k = 4$ is the order of the method. These conditions give an equation system for b_j :

$$\begin{array}{rcl} b_1 & + & b_2 = 1 \\ b_1 c_1 & + & b_2 c_2 = \frac{1}{2} \end{array}$$

which has the solution $b_1 = b_2 = \frac{1}{2}$. These c_j and b_j give the 4th order Gauss formula.

Task 3b):

The general scheme for numerical integration is

$$\hat{I}_{t_{i-1}}^{t_i}(f) = h_i \sum_{j=1}^s b_j f(t_{i-1} + c_j h_i), \quad h_i = t_i - t_{i-1}, \quad (8)$$

which is here

$$\hat{I}_0^{\pi/2} = \frac{\pi}{2} \cdot \left(\frac{1}{2} f \left(0 + c_1 \cdot \frac{\pi}{2} \right) + \frac{1}{2} f \left(0 + c_2 \cdot \frac{\pi}{2} \right) \right) = 4.3691, \quad (9)$$

with f as

$$f(t) = t \cos t + e^t. \quad (10)$$

The integral can be solved analytically:

$$\begin{aligned} I_0^{\pi/2}(f) &= \int_0^{\pi/2} t \cos t + e^t dt = [t \sin t]_0^{\pi/2} + [e^t]_0^{\pi/2} - \int_0^{\pi/2} \sin t dt \quad (11) \\ &= [t \sin t]_0^{\pi/2} + [e^t]_0^{\pi/2} + [\cos t]_0^{\pi/2} \\ &= \frac{\pi}{2} \sin \frac{\pi}{2} - 0 + e^{\pi/2} - 1 + \cos \frac{\pi}{2} - 1 \\ &= e^{\pi/2} + \frac{\pi}{2} - 2 \approx 4.3813. \end{aligned}$$

The numerical integration gives a value close to the expected one.